

Exam Date & Time: 09-Jan-2025 (02:30 PM - 05:30 PM)



# MANIPAL ACADEMY OF HIGHER EDUCATION

**Manipal Institute of Technology, Bengaluru**  
 BTech VII Semester - Make up Examination, Jan - 2025  
**SOFTWARE TESTING AND ANALYSIS [CSE-4076]**

**Marks: 50**

**Duration: 180 mins.**

## PART-A

**Answer all the questions.**

- 1) Analyse the reasons behind the requirement of software testing using a suitable example. (3)
  - A)
  - B) Identify the difference between Test Coverage and Code Coverage. (2)
  - C) For the given program, draw the data-flow graph, list the Definitions, P-Use and C-Use of all the variables used and list the DU pairs. Also give test cases to cover All-Definitions.
 

```
main()
{
int work;
double payment =0;
scanf("%d", work);
if (work > 0) {
payment = 40;
if (work > 20)
{
if(work <= 30)
payment = payment + (work - 25) * 0.5;
else
{
payment = payment + 50 + (work -30) * 0.1;
if (payment >= 3000)
payment = payment * 0.9;
}
}
}
printf("Final payment", payment);
}
```

 (5)
- 2) A function named compute-electricity-bill was written to compute the electricity bill by an electricity distribution company. This function takes two parameters, the number of units consumed by a customer and the corresponding customer type. The customer type (4)

- A) is an integer value in the range 1 to 5 indicating whether the customer is domestic, industrial, commercial establishment, etc. The tariff depends not only the customer type, but also on the number of units consumed. The slabs for different charges based on the units consumed are 0 to 100 units, 100 to 200 units, 200 to 500 units, and 500 units and above. Identify the test cases for weak normal and weak robust equivalence class testing.

- B) Consider the following program:

```
main ( )
{
int a,b,c, sum, diff, mul;
scanf( "%d %d %d", &a, &b, &c);
sum=calsum(a,b,c);
diff=caldiff(a,b,c);
mul=calmul(a,b,c);
printf("%d %d %d", sum,diff,mul);
}
calsum(int x, int y, in z){
int d;
d=x+y+z; return(d); }
```

(3)

(i). Suppose, the main ( ) module is not ready for the testing of calsum ( ) module. Apply suitable testing strategy to test the whole system in such a scenario.

(ii). Suppose modules caldiff() and calmul( ) are not ready when called in main().Apply suitable testing strategy to test the whole system in such a scenario.

- C) Construct the control flow graph for the given program.

```
begin
int x,y;
int z;
input(x,y):z=0;
if(x< 0 and y< 0)
{
z=x*x;
if(y>=0)
z=z+1;
}
else
Z=x*x*x;
Output(z)
}
end
```

(3)

Identify the feasible and Infeasible path from the control flow graph.

- 3) List different types of interface errors.

(2)

- A)
- B) Analyse different coverage criteria with respect to call graphs in Integration Testing. (3)
- C) Consider the following program.

```

public class TrashAndTakeOut
{
    public static void main (String []argv)
    {
        int x = 0;
        x = Integer.parseInt (argv[0]);
        trash (x);
    }
    public static void trash (int x)
    {
        int m, n;
        m = 0;
        if (x > 0)
            m = 4;
        if (x > 5)
            n = 3*m;
        else
            n = 4*m;
        int o = takeOut (m, n);
        System.out.println ("o is: " + o);
    }
    public static int takeOut (int a, int b)
    {
        int d, e;

        d = 42*a;
        if (a > 0)
            e = 2*b+d;
        else
            e = b+d;
        return (e);
    }
}

```

(5)

- (i) Identify Last Defs and First Uses of different variables.
- (ii) Identify Coupling du-pairs in triplet form (method name, variable name, statement number)

- 4) Analyse different regression test selection process. (3)

- A)
- B) An airline offers only flights to India and Asia. Under special conditions, a discount is offered on the normal airfare: (5)

- o Passengers older than 18 with destinations in India are offered a discount of 20%, if the departure is not on a Monday or Friday.
- o For destinations outside of India, passengers are offered a discount of 25%, if the departure is not on a Monday or Friday.
- o Passengers who stay at least 6 days at their destination receive an additional discount of 10%.
- o Passengers older than 2 but younger than 18 years are offered a discount of 40% for all destinations.
- o Children 2 and under travel for free

Design test cases for this program using decision table testing.

- C) Analyse the advantage of MC/DC coverage over condition coverage using a suitable example. (2)
- 5) Consider the following program P and three mutants

- A)
- ```

1. main(argc, argv){
2. int argc, r, i;
3. char *argv[ ];
4. r = 1;
5. for i = 2 to 3 do
6. if (atoi(argv[i]) > atoi(argv[r])) r = i;
7. printf("Value of the rank is %d n", r);
8. exit(0); }
```
- (5)

M1: Change line 5 to for i = 1 to 3 do

M2: Change line 6 to if (i > atoi(argv[r])) r = i;

M3: Change line 6 to if (atoi(argv[i]) ≥ atoi(argv[r])) r = i;

Design minimum number test cases to achieve 100% adequacy with respect to mutation testing. With a neat table, show the output of P and mutants for each test case.

- B) Consider a program that takes three numbers as input and print the values of these numbers in descending order. Its input is a triple of positive integers (say x, y and z) and values are from interval [300,700]. Generate normal and robust boundary value test cases. (5)

-----End-----